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MODULE - V

Induction Generator – Principle,

Grid connected & Self excited operation,

Torque-Slip characteristics of Induction Machine.

Single phase Induction motor – Double field revolving theory

Equivalent circuit, Torque – Slip Curve

Types of Single phase Induction motor

- Split phase,
- Capacitor start,
- Capacitor start & run type,
- Shaded pole Induction motor,

Applications

Induction Generator

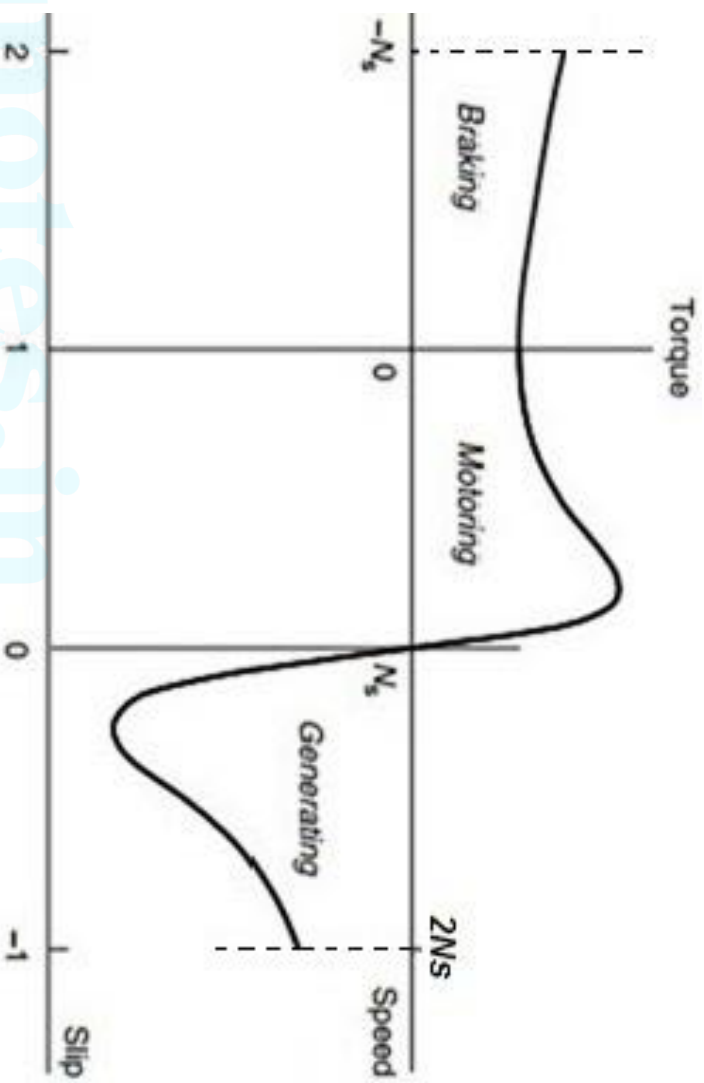
- An Induction Generator is a poly phase Induction machine working as a generator.
- When an Induction machine is connected to 3 phase AC supply, it will work as a motor drawing power from supply
- The motor speed will be less than synchronous speed & slip is positive.
- If the induction motor is coupled to a prime mover (whose speed can be controlled), the speed of Induction motor can be increased above synchronous speed
- Now slip is negative & direction of current, power & torque reverses.
- i.e, Machine works as a generator & produces electrical energy
- The produced electrical energy is fed back to supply.
- The Torque – Speed characteristics of Induction generator is shown in figure.
- The Induction generator takes the required reactive power from supply to produce the magnetic field in machine. So it is not a self excited machine.

➤ In an Induction generator, the magnitude of voltage & frequency of generated AC voltage depends on the magnitude of voltage & frequency of stator supply

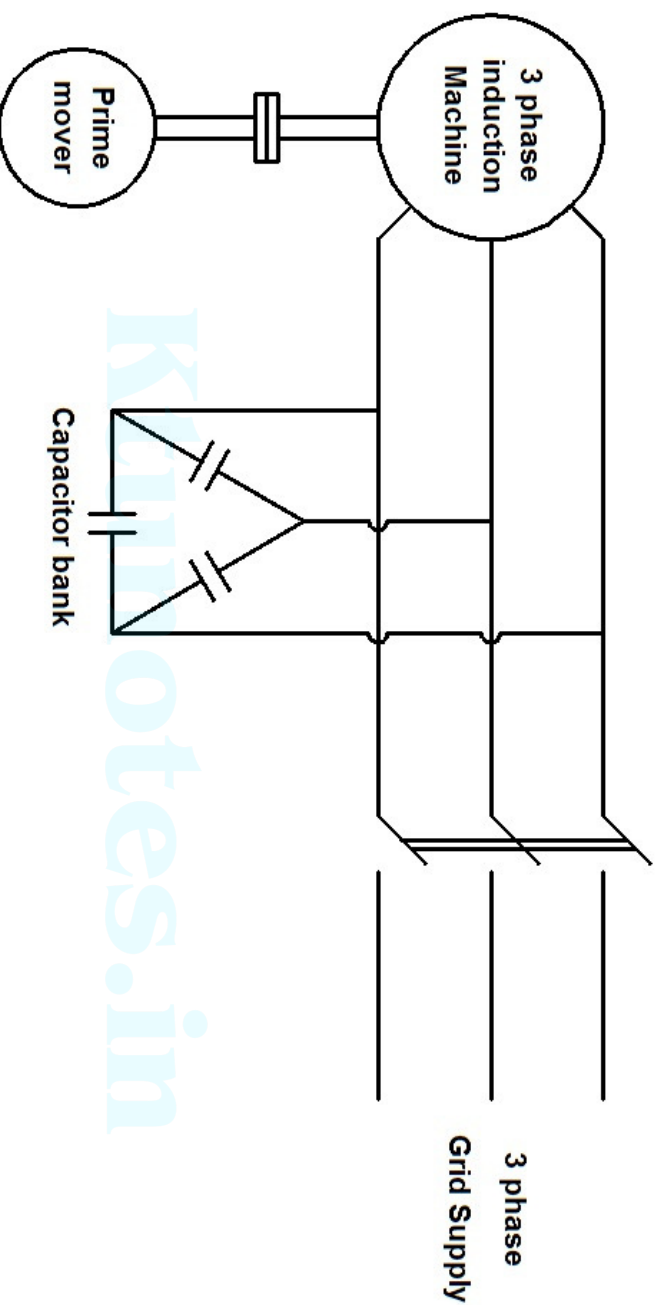
➤ An Induction generator has 2 modes of operation,

1. Grid connected operation

- Here the Induction generator is directly connected to grid as shown in figure
- The reactive power required for machine to produce magnetic field is drawn from grid supply
- The generator is started as an Induction motor & then speed is increased above synchronous speed by using the prime mover to work as a generator

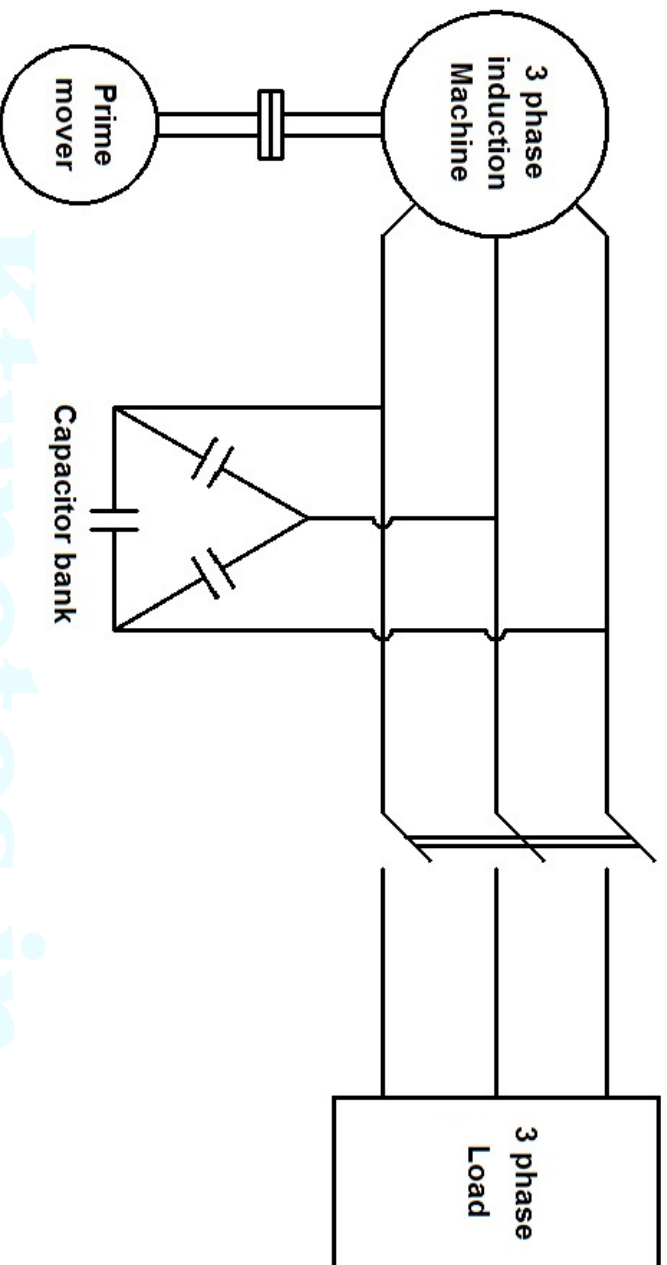


- To avoid drawing of reactive power from supply, usually a capacitor bank is connected as shown in figure



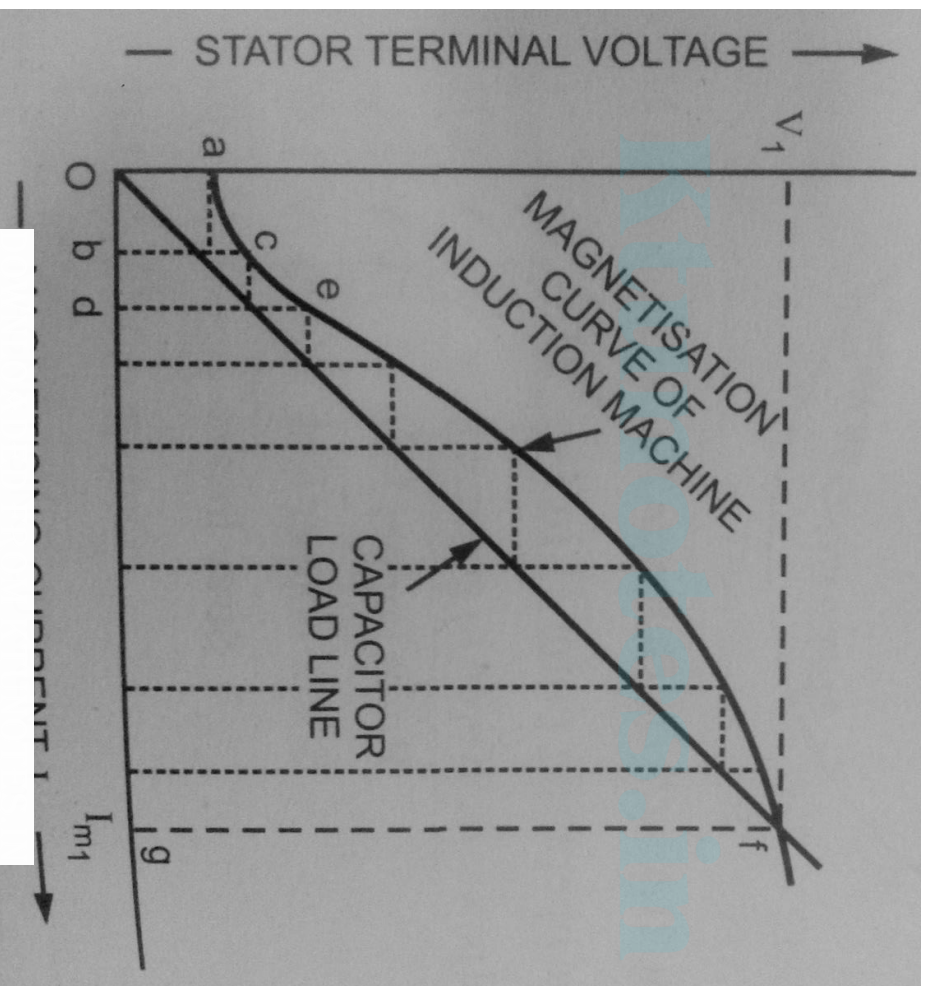
2. Self excited/Stand alone operation

- Here Induction generator does not require an existing AC supply for obtaining reactive power to produce magnetic field
- In self excited system, a capacitor bank is connected across the Induction machine stator terminals as shown in figure

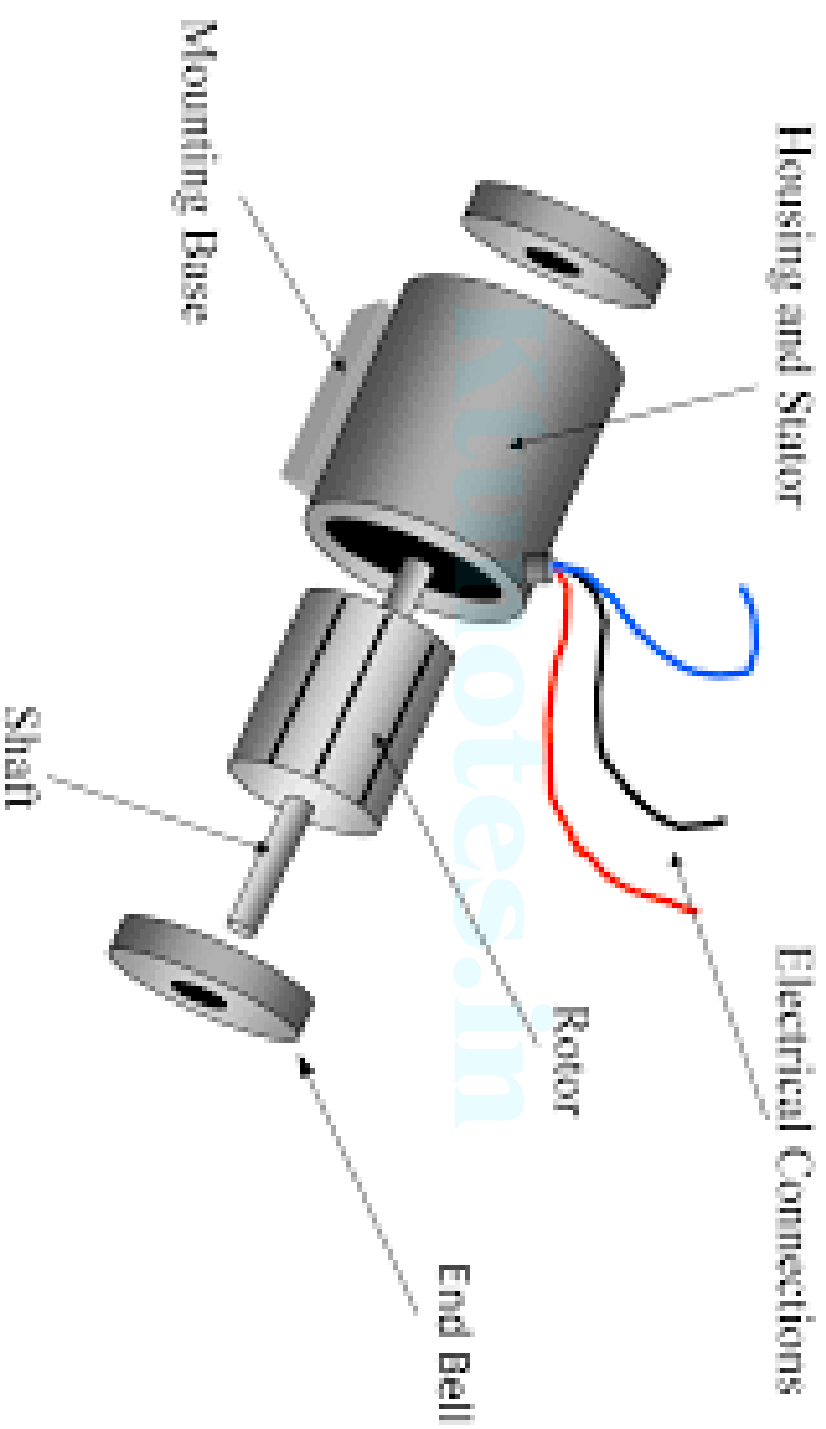


- When Induction generator is rotated at synchronous speed, due to residual magnetism in rotor circuit, a small voltage induces in stator
- This voltage produces a capacitor current. This capacitor current flows to stator & produces a flux in the stator which aids the residual flux
- As a result total flux increases & stator induced voltage also increases
- Now capacitor bank produces much larger current as a result of this increased voltage

- This will continue as a cumulative process of voltage build up till the saturation curve of Induction generator intersects capacitor load line at the point 'f', thus giving no load generated voltage of V_1 for magnetizing current of I_m
- Here higher the value of capacitor, greater will be the voltage build up



SINGLE PHASE INDUCTION MOTOR



- ✓ The single-phase induction machine is the most frequently used motor for domestic as well as industrial low power application.
- ✓ These motors are also known as fractional KW motor s because most of these motors are constructed in fractional kilo-watt capacity.
- ✓ Single phase induction motors require just one power phase for their operation.
- ✓ The single phase motors are widely seen in many applications as
 - it is simple in construction,
 - cheap in cost,
 - reliable and
 - easy to repair and maintain.
- ✓ Due to these advantages, the single phase motor is used in vacuum cleaner, fans, washing machine, centrifugal pump, blowers, etc.

X The main disadvantage of single phase induction motor are low output, low overload capacity, low power factor and also low efficiency compared to three phase induction motor.

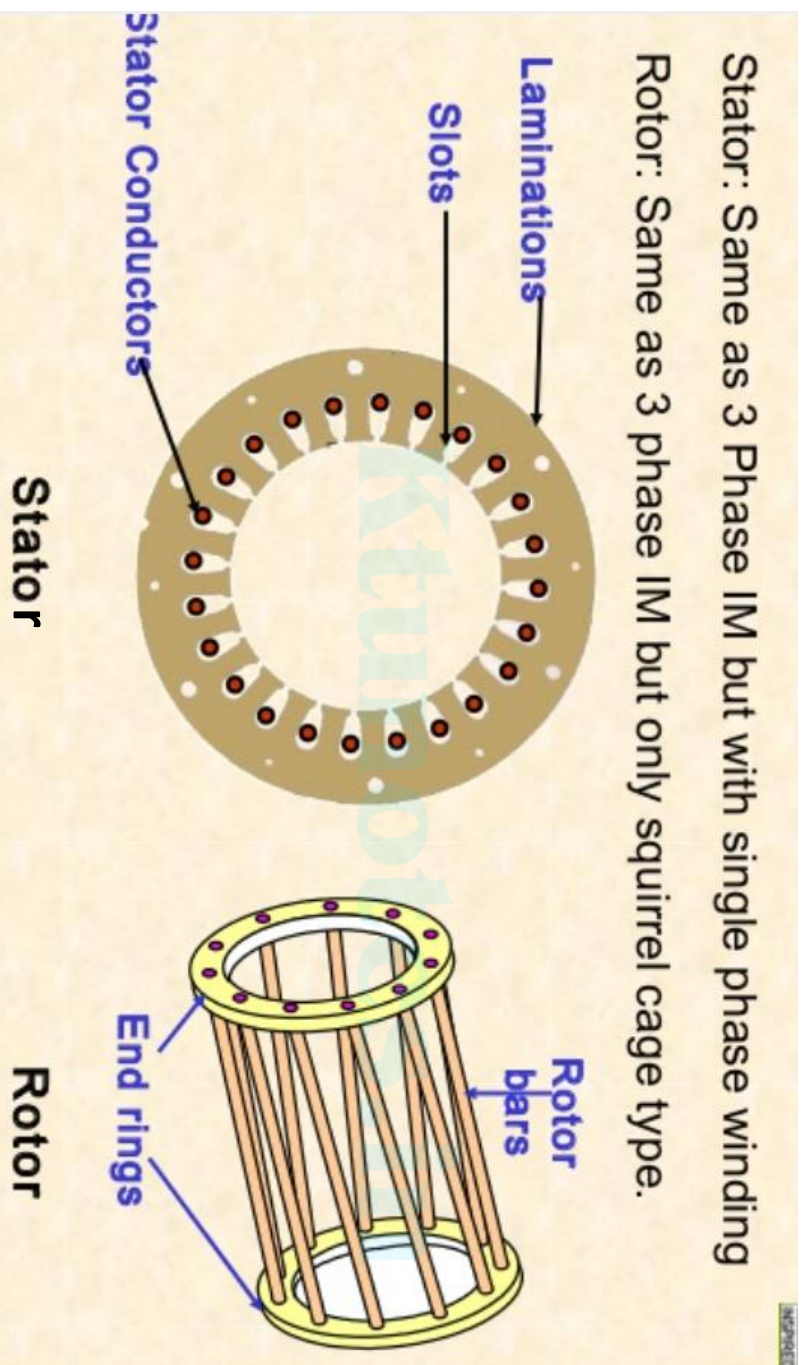
X Also a single phase induction motor is not self starting.

- An additional winding known as starting winding or auxiliary winding is also provided with stator to make the motor self starting.
- So it can be temporarily converted into a two-phase motor while starting.
- These two windings are spaced 90° electrical degrees apart and are connected in parallel across a single phase supply.
- The Phase difference of 90° degree is obtained by connecting a capacitor in series with the starting winding.

Construction of Single Phase Induction Motor

Stator: Same as 3 Phase IM but with single phase winding

Rotor: Same as 3 phase IM but only squirrel cage type.



The stator has two windings instead of one three-phase winding in [three phase induction motor](#).

PRINCIPLE OF OPERATION

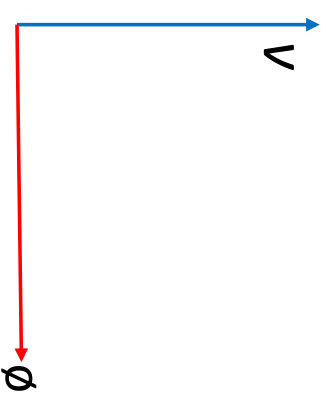
- When a single phase AC supply is connected to stator winding, an alternating flux is produced in stator.
- This alternating flux which is pulsating in nature also links with the rotor conductors and hence the rotor conductors cut the magnetic flux.
- According to the Faraday's law of electromagnetic induction, emf gets induced in the rotor causes induced current in the rotor bars.
- This induced current produce alternating rotor flux which opposes the stator field as per Lenz's Law.
- So a single phase induction motor is not self starting.
- However, if the rotor is given a initial start by external force in either direction, it develops an initial torque then motor accelerates to its final speed and keeps running with its rated speed.
- Thus single phase induction motor is not self starting and need special design to make it self starting.

Double-field Revolving Theory

An alternating sinusoidal flux can be represented by two revolving fluxes, each equal to half the value of the alternating flux and each rotating synchronously (Ns = 120f/P) in opposite direction*

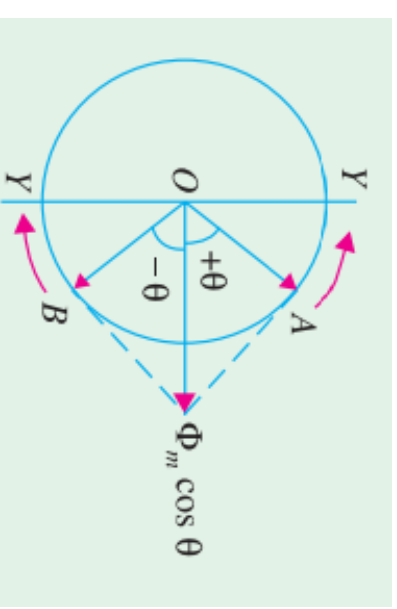
$$v = V_m \sin(\theta)$$

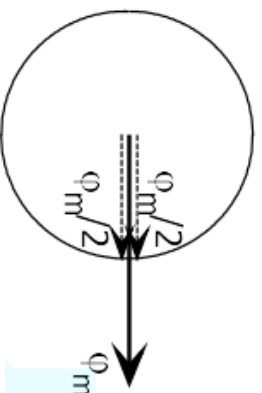
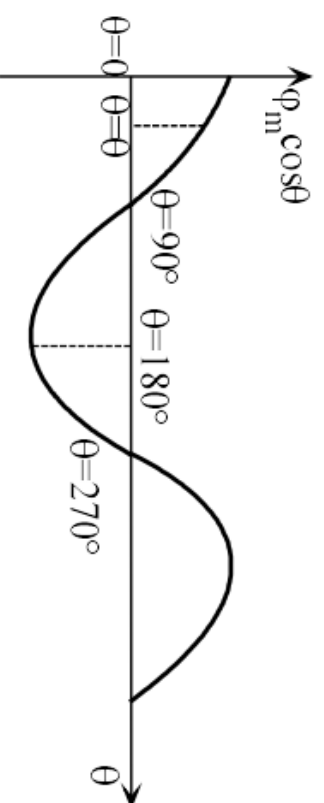
$$\phi = \phi_m \cos(\theta)$$



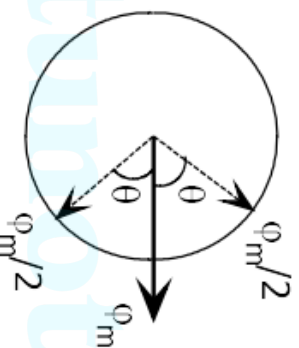
$$\phi = \frac{\phi_m}{2} \cos(\theta) + \frac{\phi_m}{2} \cos(\theta)$$

$$\phi = \frac{\phi_m}{2} \cos(\theta) + \frac{\phi_m}{2} \cos(-\theta) \dots\dots\dots(1)$$

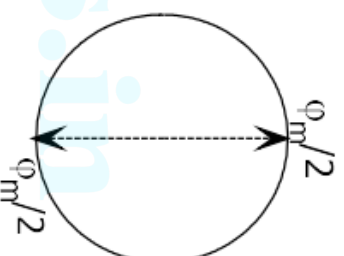




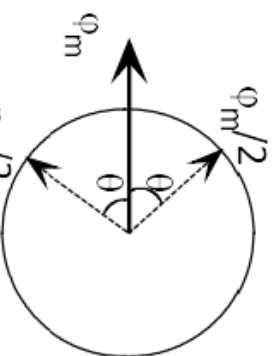
$\theta = 0$



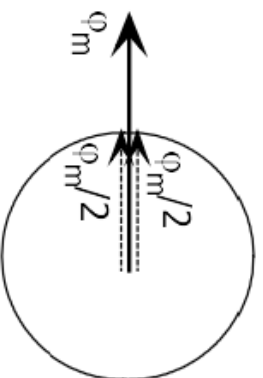
$\theta = 0$ (any angle)



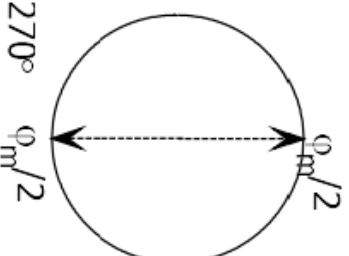
$\theta = 90^\circ$



$\theta = 180 - 0$



$\theta = 180^\circ$



$\theta = 270^\circ$

If the slip of the rotor is s with respect to the forward rotating flux (i.e. one which rotates in the same direction as rotor) then its slip with respect to the backward rotating flux is $(2 - s)^*$.

Each of the two component fluxes, while revolving round the stator, cuts the rotor, induces an e.m.f. and this produces its own torque. Obviously, the two torques (called forward and backward torques) are oppositely-directed, so that the net or resultant torques is equal to their difference.

Now, power developed by a rotor is $P_g = \left(\frac{1-s}{s} \right) I_2^2 R_2$

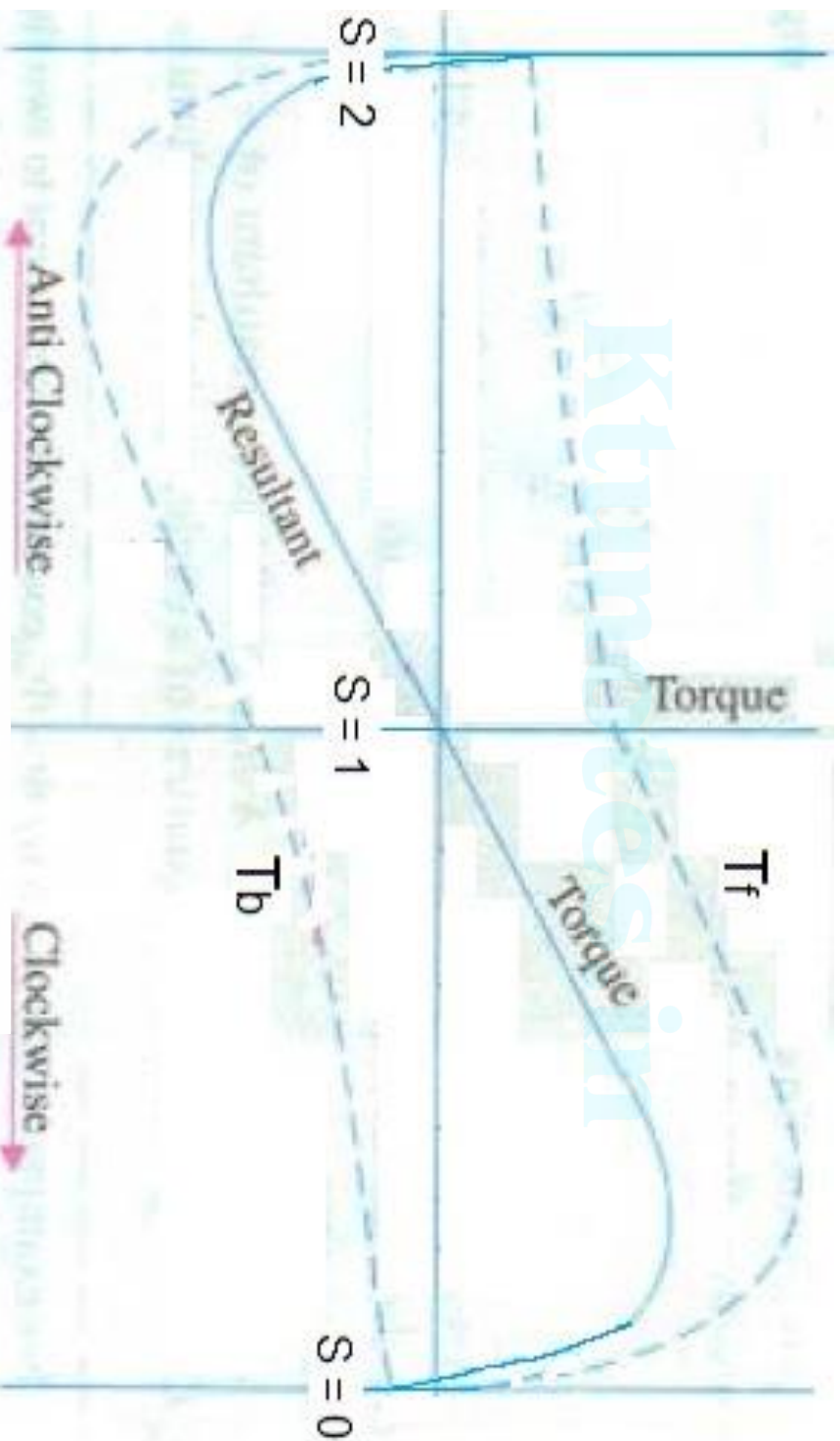
If N is the rotor r.p.s., then torque is given by $T_g = \frac{1}{2\pi N} \cdot \left(\frac{1-s}{s} \right) I_2^2 R_2$

Now, $N = N_s (1-s) \quad \therefore T_g = \frac{1}{2\pi N_s} \cdot \frac{I_2^2 R_2}{s} = k \cdot \frac{I_2^2 R_2}{s}$

Hence, the forward and backward torques are given by

$$T_f = K \frac{I_2^2 R_2}{s} \quad \text{and} \quad T_b = -K \cdot \frac{I_2^2 R_2}{(2-s)}$$

- The resultant maximum torque is less than in a 3 phase Induction motor
- The resultant torque become zero at a speed slightly below synchronous speed
- At synchronous speed, torque is negative



$$\text{Total torque} = T = T_f + T_b$$

At standstill,

$$s = 1 \text{ and } (2 - s) = 1.$$

$$\text{Hence, } T_f = -T_b$$

$$\text{Total torque} = 0$$



Thus single phase induction motor is not self starting

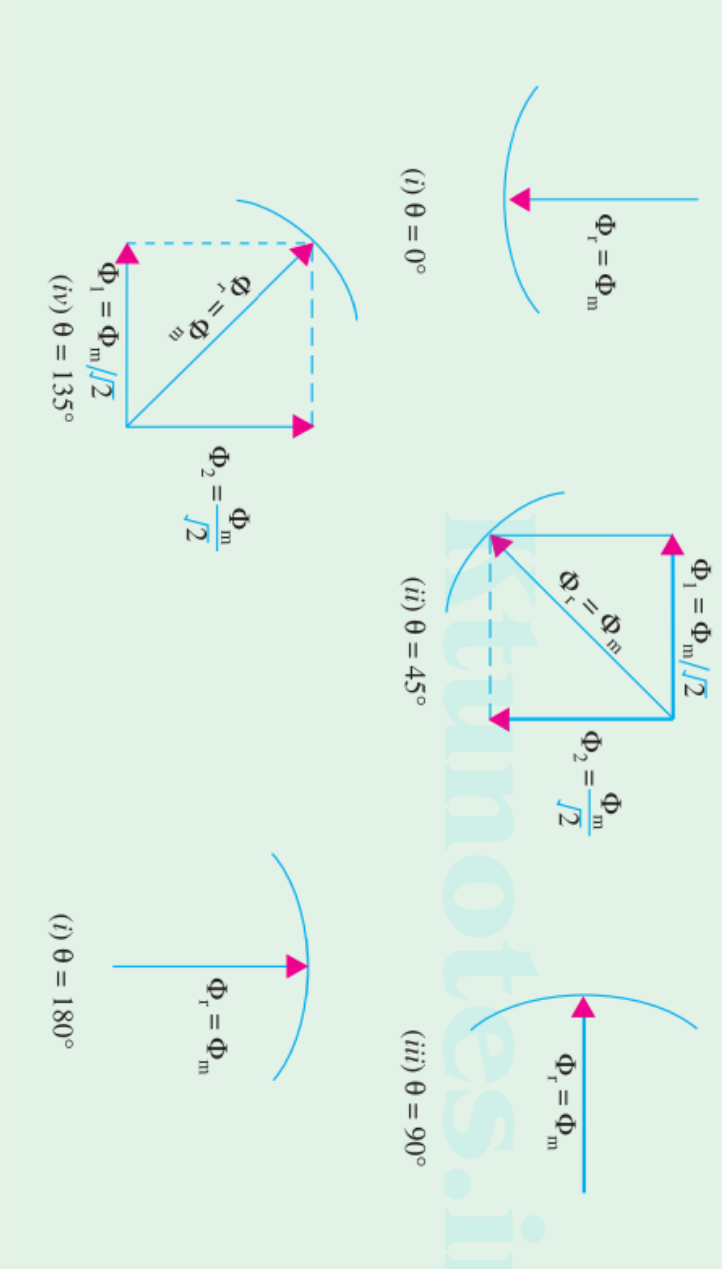
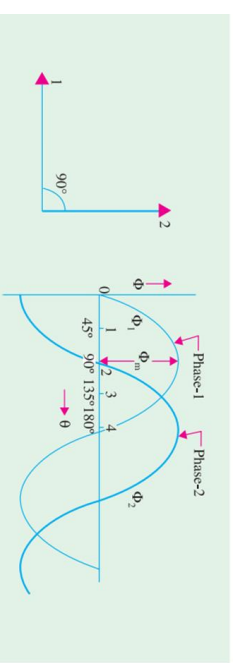
However, if the rotor is started somehow, say, in the clockwise direction, the clockwise torque starts increasing and, at the same time, the anticlockwise torque starts decreasing. Hence, there is a certain amount of net torque in the clockwise direction which accelerates the motor to full speed.

METHODS FOR MAKING SINGLE PHASE INDUCTION AS SELF STARTING MOTOR

- It is temporarily converted into a two-phase motor during starting period.
- The stator of a single-phase motor consists of an extra winding, known as starting (or auxiliary) winding, in addition to the main or running winding.
- These two windings are connected in parallel across the single-phase supply. These two windings are having equal number of turns and placed at a space angle of 90° (electrical)
- When the stator winding is wound for two phase, stator flux becomes rotating type, rather than alternating type. This would result in an unidirectional torque.
- Then the induction motor will become self starting.

PRODUCTION OF ROTATING FIELD

Consider two-phase Supply



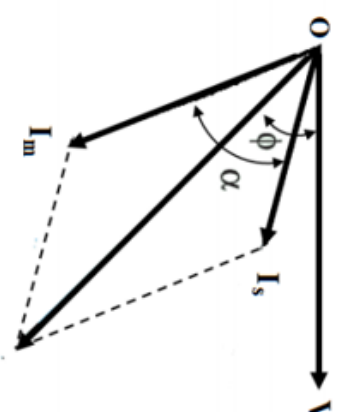
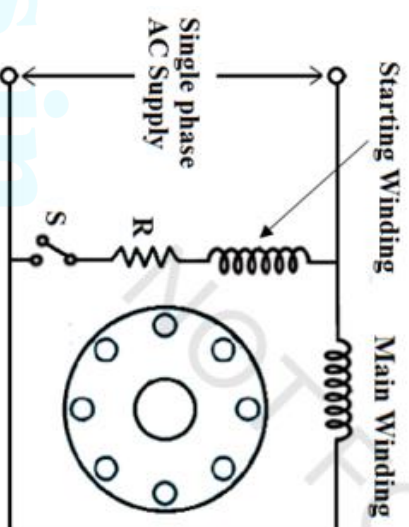
1. The magnitude of the resultant flux is constant and is equal to Φ_m — the maximum flux due to either phase.
2. The resultant flux rotates at synchronous speed given by $N_s = 120 \text{ f/P rpm}$.

Making 1 phase Induction motor self starting

- 1 phase Induction motor is not self starting & if the rotor is rotated in one direction by some external means, it will continue to run in the direction of rotation
- This method of starting is undesirable & require some method for creating a rotational magnetic field & make the motor self starting.
- This can be achieved by converting 1 phase supply into 2 phase supply by providing an additional winding called auxiliary/starting winding
- Once the motor has achieved sufficient speed, the additional windings can be disconnected & the motor will continue to rotate
- Depending on the method employed for making the motor self starting, 1 phase Induction motors are classified into
 - a) Split phase motor – started by 2 phase motor action through the use of a starting winding
 - b) Capacitor motor - started by two-phase motor action through the use of an auxiliary winding and a capacitor
 - c) Shaded pole motor - started by the motion of the magnetic field produced by means of a shading coil around a portion of the pole structure

a) SPLIT PHASE MOTOR

- It is provided with two windings – *main or running winding (M)* and *auxiliary or starting winding (S)*
- The main winding is permanently connected to the supply and the starting winding is placed 90° electrical from main winding and is connected to the supply only during starting through a centrifugal switch.
- The *main winding has low resistance and high reactance* and the *auxiliary winding has high resistance and low reactance*
- So the *current flowing through these windings has a phase difference α (25° to 30°)* and the flux produced by these currents also has a phase difference α .

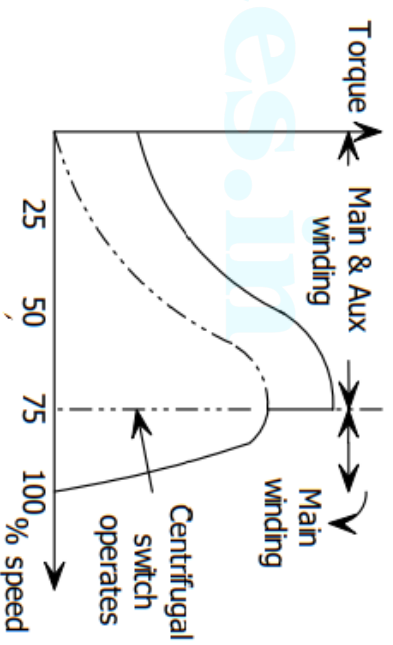


- *The resultant flux will be rotating one.*
- *Starting torque is proportional to $\sin \alpha$*
- Split phase induction motor is having moderate starting torque and are used in washing machines, blowers, grinders etc.

These motors are often used in preference to the costlier capacitor-start motors.

Typical applications are :

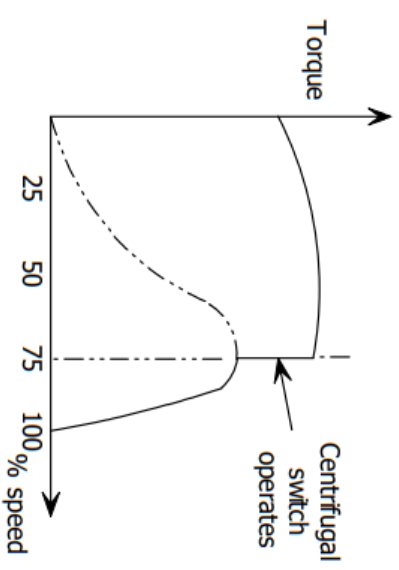
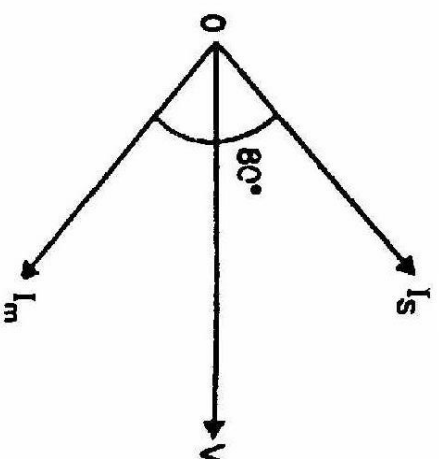
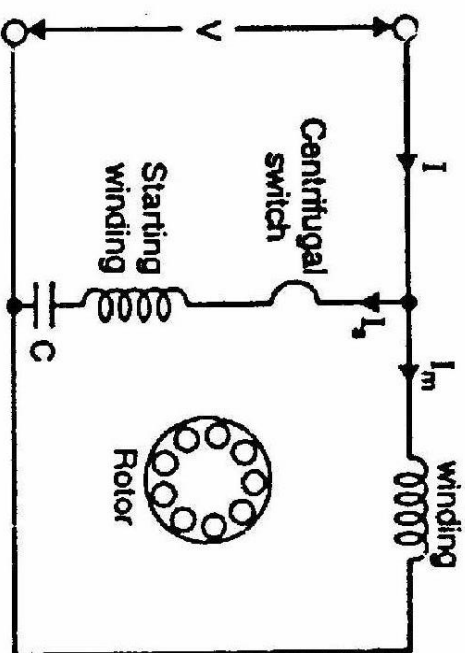
1. Fans and blowers,
2. Centrifugal pumps
3. Separators
4. washing machines
5. small machine tools



torque/speed characteristic

b) Capacitor motor

- In split phase motor, the maximum phase difference between starting and running winding current is 30°
- To improve starting torque (by increasing the angle), a capacitor is included in the starting winding circuit.
- Depending on the connection of capacitor, these motors are classified into
 - ❖ i) **Capacitor start motor**
- Here in addition to split phase motor, a capacitor is connected in series with the starting winding
- The value of capacitor is chosen such that the phase difference ' α ' is about 80°
- Used in applications requiring high starting torque like refrigerator, air conditioner, compressors etc



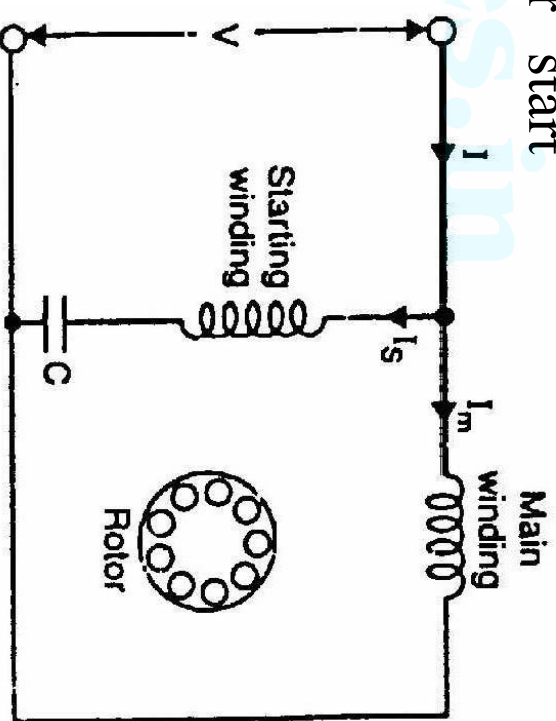
torque/speed characteristic

ii) Capacitor start capacitor run motor

- This motor is identical to Capacitor start motor
- The only difference is that both starting and running winding remain connected to supply during starting and running.
- Two designs are generally used

Design 1

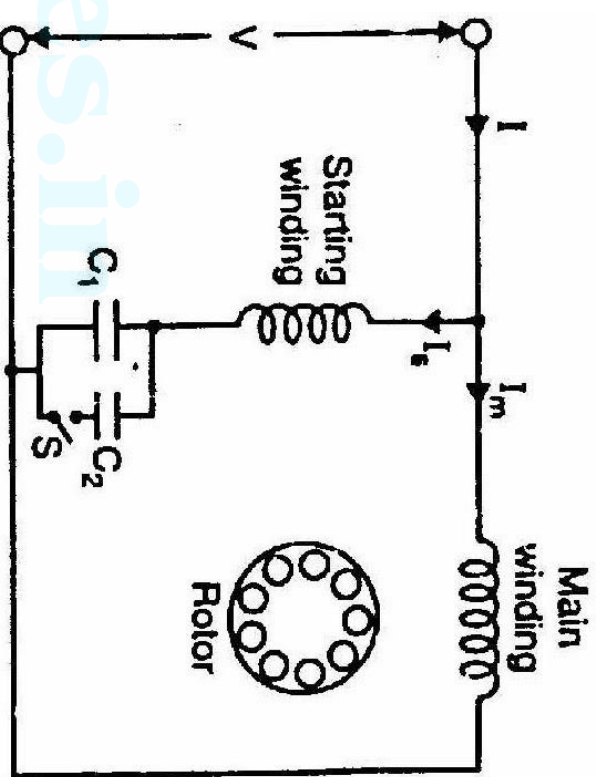
- Here a single capacitor is used for both starting and running



- No need of centrifugal switch
- Got improved power factor and efficiency

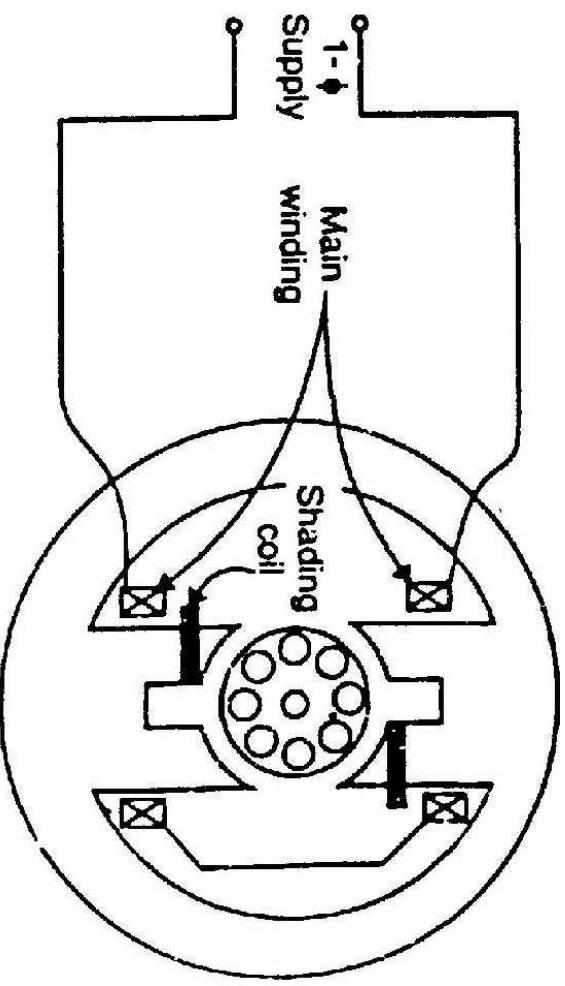
Design 2

- Here **two capacitors C_1 and C_2** are connected in series with starting winding.
- The smaller capacitor **C_1** required for optimum **running** condition is permanently connected in series with starting winding.
- The larger capacitor **C_2** act as **starting** capacitor and is connected in parallel to C_1 through a centrifugal switch.
- This motor produces **a constant torque**.
- These motors are mostly used in hospitals, studios and in other places where silence is important.



c) Shaded pole motor

- The shaded-pole motor has a cage rotor with salient poles in the stator.
- On one side of each pole a slot is cut and a **copper shading ring** is embedded in the slot.
- The stator supply current produces an alternating flux which induces a current in shading ring

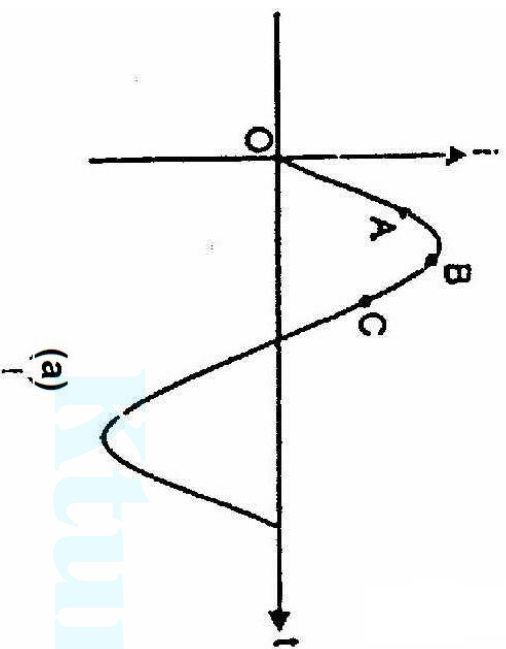


Working

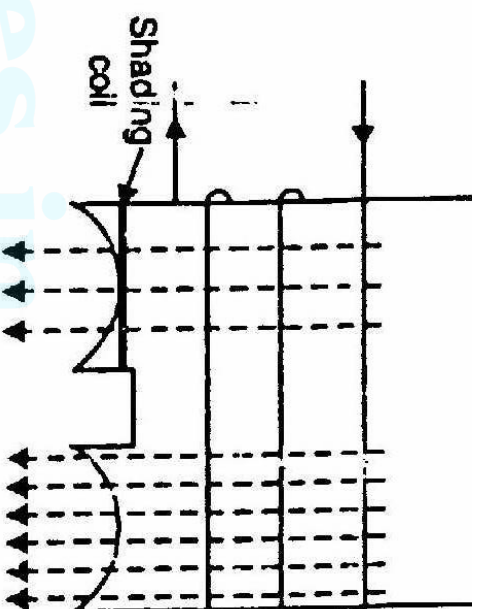
The operation of the motor can be understood by considering one pole of stator with shading ring as shown in figure

- During the portion OA of the alternating-current cycle [See Fig. (a)], the flux begins to increase and an EMF is induced in the shading coil. The resulting current in the shading coil will be in such a direction (Lenz's law) so as to counter the change in flux.

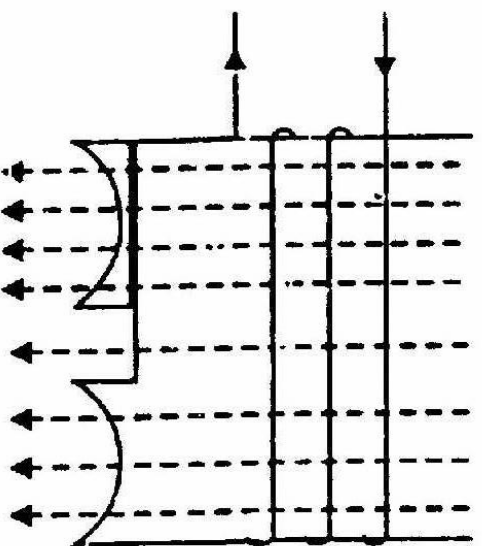
Thus the flux in the shaded portion of the pole is weakened while that in the unshaded portion is strengthened as shown in Fig. (b).



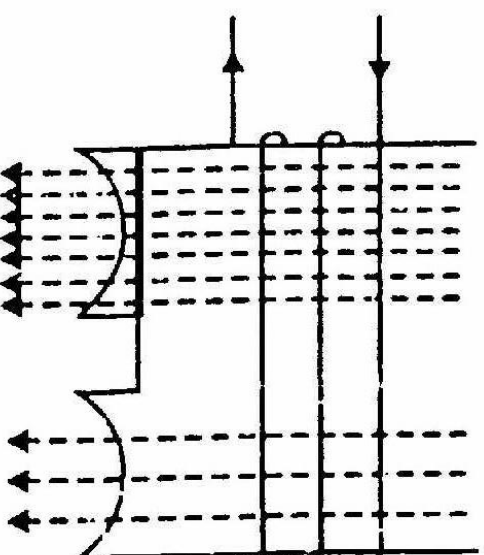
(a)



(b)



(c)



(d)

- ii) During the portion AB of the alternating-current cycle, the flux has reached almost maximum value and is not changing. Consequently, the flux distribution across the pole is uniform [See Fig. (c)] since no current is flowing in the shading coil
- iii) As the flux decreases (portion BC of the alternating current cycle), current is induced in the shading coil so as to oppose the decrease in current. Thus the flux in the shaded portion of the pole is strengthened while that in the unshaded portion is weakened as shown in Fig. (d)
 - This shifting flux is like a rotating field moving in the direction from unshaded portion to the shaded portion of the pole.
 - The rotor is of the squirrel-cage type and is under the influence of this moving field.
 - Consequently, a small starting torque is developed.
 - As soon as this torque starts to revolve the rotor, additional torque is produced by single phase induction motor action.
 - The motor accelerates to a speed slightly below the synchronous speed and runs as a single phase induction motor

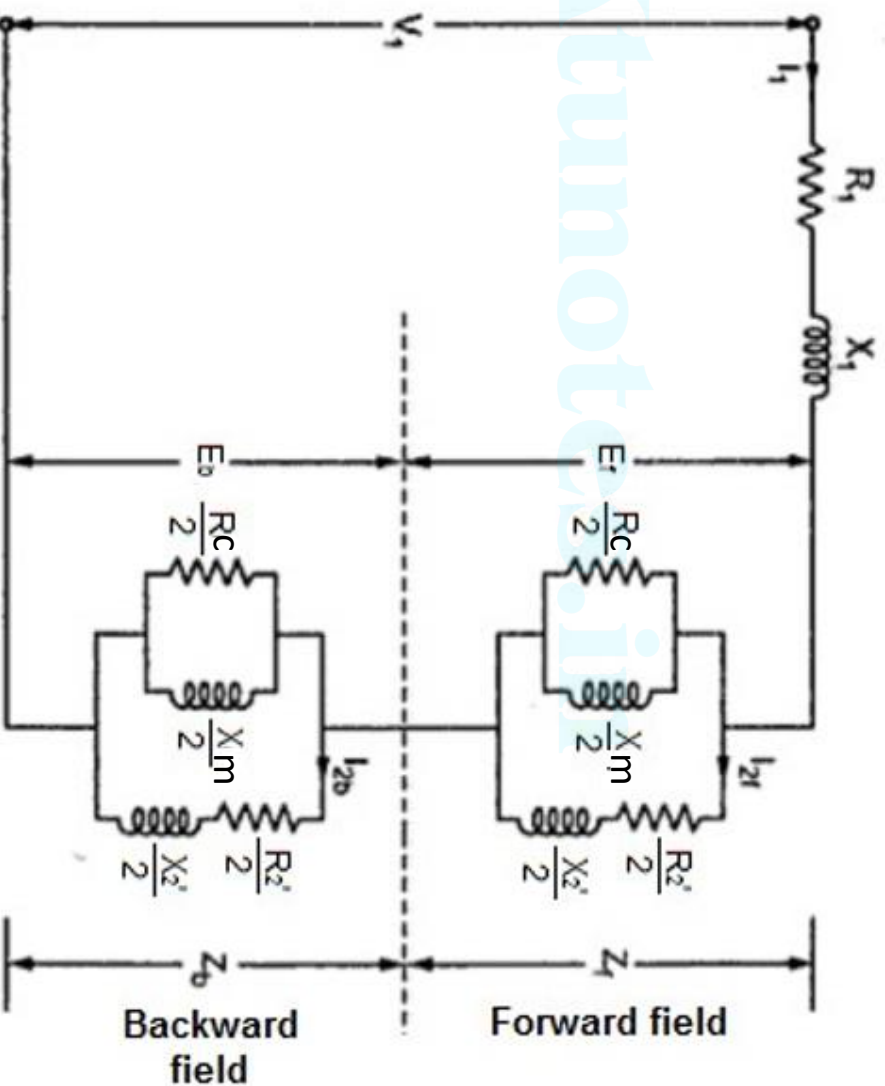
- ✓ The salient features of this motor are extremely simple construction and absence of centrifugal switch.
- ✓ Since starting torque, efficiency and power factor are very low, these motors are only suitable for low power applications e.g., to drive: (a) small fans (b) toys (c) hair driers (d) desk fans etc.
- ✓ The power rating of such motors is upto about 30 W.

Equivalent circuit of 1 phase Induction motor

- When the stator of a single-phase induction motor is connected to single-phase supply, the stator current produces a pulsating flux.
- According to Double-field revolving theory, this flux can be resolved into 2 flux components having same magnitude & revolving in opposite directions at the synchronous speed.
- Each of these fluxes induces currents in the rotor circuit and produces induction motor action similar to that in a 3-phase induction motor.
- Therefore, a single-phase induction motor can be imagined to be consisting of two motors, having a common stator winding but with their respective rotors revolving in opposite directions.
- Each rotor has resistance and reactance half the actual rotor values
 - Let R_1 = resistance of stator winding
 - X_1 = leakage reactance of stator winding
 - X_m = total magnetizing reactance
 - R_c = total resistance equivalent to core losses
 - R_2' = resistance of the rotor referred to the stator
 - X_2' = leakage reactance of the rotor referred to the stator

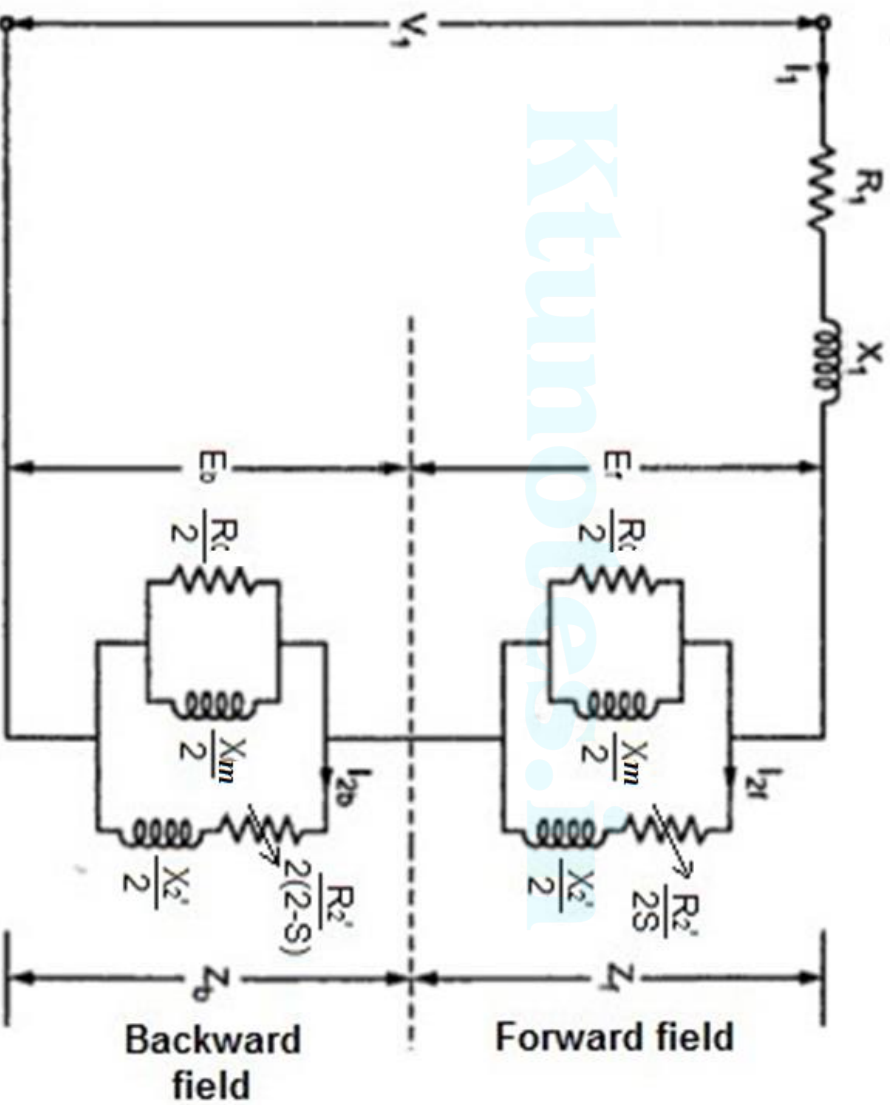
❖ At standstill

- At standstill, the motor is simply a transformer with its secondary short circuited.
- Therefore, the equivalent circuit of single phase motor at standstill will be as shown in Figure



❖ Rotor running

- Now consider that the motor is rotating at some speed in the direction of the forward revolving field, the slip being s .
- The equivalent circuit under this situation is shown in figure



SELECTION OF AC MOTORS FOR DIFFERENT APPLICATIONS:

S No	Application	Requirement	Motor
1	Textile mill - Ginning	Constant speed from 200 to 1500rpm	3-phase squirrel cage IM
	Textile mill - Cording	High starting torque, high overload capacity, 600 or 700rpm constant operating speed	3-phase slip ring IM
	Textile mill - Looms	Frequent starts and stops, constant speed operation, high starting torque	3-phase squirrel cage IM
2	Cranes	Accurate control of acceleration and retardation; wide speed control	3-phase slip ring IM
3	Steel mills	High starting torque, high overload capacity, frequent starts and stops, wide range of speed control	3-phase slip-ring iM
4	Sugar mills	500 to 1500 rpm; high starting torque	Two speed pole changing 3-phase squirrel cage motor
5	Compressors	Controlled acceleration, 700 to 1000rpm	3-phase IM or synchronous motor with frequency control
6	Pumps and blowers	Speed range 1:5	3-phase squirrel cage or slip-ring IM
7	Cement mills	Small starting current, high overload capacity, frequent starts and stops	3-phase slip ring IM with rotor resistance starting or synchronous motor or squirrel cage IM with frequency control
8	Domestic fans and pumps		1-phase induction motors